



Figure 10: Effect of transferring steering vector from Llama 2 7B base model to chat model. Lines correspond to different behaviors.

Another compelling aspect of CAA is its compatibility with standard alignment techniques such as system-prompting and finetuning. The additive nature of CAA’s steering capabilities allows for a layered approach to model steering, where CAA can refine and adjust model outputs further, even after applying other alignment methods.

The ability of CAA to control latent variables within the model’s internal state opens up new avenues for inference-time control. It has high sample efficiency and strong generalization, and is particularly advantageous in scenarios requiring the precise modulation of model behavior or the elicitation of internal states that are difficult to trigger with prompting alone.

Moreover, the insights gained from applying CAA extend beyond immediate practical benefits, offering a deeper understanding of models’ internal representation and processing of high-level concepts and shedding light on the emergence of linear representations.

In conclusion, by enabling precise, efficient, and effective control over model behavior, CAA contributes to the broader goal of creating AI systems that are controllable and aligned with human values and provides additional insights into emergent linear representations of abstract concepts in LLMs.

9.1 Suggested future work

Steering at targeted token positions

Our intervention applies the steering vector at every token position after the user’s prompt. This results in a cap on the amount by which we can perturb the representations before degrading text quality. By intervening at a smaller, more targeted, subset of tokens, a better trade-off between intervention size and effect size may be achieved.

Steering outside the residual stream

CAA could be applied at other points in the model, such as after the MLP but before merging into the residual stream. By intervening in other positions, we could learn more about where representations are localized in the model and achieve more targeted effects.

Application to red-teaming

Validating if finetuning and RLHF have made models robustly safe is challenging. Although these methods reduce the likelihood of specific dangerous outputs, unwanted behaviors can often still be elicited with adversarial or unusual inputs. For example, users can often find “jailbreaks” to make LLMs output harmful content. However, systematically finding inputs that reveal these flaws is challenging. CAA could be used as an adversarial intervention to trigger unwanted behaviors in models more efficiently (Panickssery, 2023a). If a behavior can be easily triggered through techniques such as CAA, it may also occur in deployment. Conversely, the inability to elicit behaviors via small internal perturbations could serve as a stronger guarantee of safety.

10 Limitations

GPT-4 eval

While GPT-4 provides a scalable way to evaluate open-ended generation, it has some limitations. GPT-4’s scores can be sensitive to scoring prompt wording, potentially introducing noise in the evaluations. In addition, LLMs have their own biases that could cause systematic differences from human evaluators. To mitigate these limitations, we manually inspect a sample of GPT-4’s ratings to check for surprising results that are inconsistent with our own manual scores, and find that these correspond well, in line with other work such as [Hackl et al. \(2023\)](#) which finds GPT-4 to be a consistent and reliable rater. We also sample many open-ended generations with the steering intervention and see a consistent noticeable change in the direction being steered, with preservation of subjective text quality, as demonstrated in the samples seen in Appendix G.

Finetuning baseline optimization

When comparing to finetuning, we do not optimize supervised finetuning hyperparameters such as learning rate, number of epochs, or precise loss function. The set of hyperparameters initially chosen achieves high accuracy (>90%) on most of the test sets. However, better results can be achieved with more optimization, resulting in a smaller effect size for CAA on top of finetuning. A possible modification to the finetuning intervention is using a contrastive loss function that penalizes selecting the negative answer rather than just incentivizing the selection of the positive answer.

Prompting baseline optimization

We test several system prompt options and few-shot prompting setups when constructing the comparison to the prompting baseline. However, it is challenging to search over all possible prompting interventions. It is possible that better steering effects could be achieved via prompting alone if more effort were applied to finding effective prompts. However, this indicates that CAA is a more reliable steering method as it does not require manual prompt optimization.

Vector normalization choices

CAA steering vectors resulting from our datasets have different norms. We normalize steering vector magnitudes across all behaviors to standardize

across behaviors before applying steering multipliers. However, an additional axis of norm variation is the norm over layers. The residual stream norm generally grows exponentially over the forward pass ([Heimersheim and Turner, 2023](#)), so we choose not to normalize over the layers to preserve a “natural norm” given the sampled activations. However, this could skew our result for layer optimality as we do not search over different multipliers per layer. Different magnitudes could be optimal for different layers. In contrast, our approach to CAA hyperparameter search involves first finding an optimal layer using a *constant* multiplier and then testing a range of multipliers at the resultant best layer.

Ethics Statement

Our method aligns with the goal of making AI systems more helpful, honest, and harmless. By enabling precise steering of language model outputs, CAA contributes to reducing risks associated with misaligned or unsafe behaviors, thereby enhancing the safety and reliability of AI systems. We are aware of the potential for misuse of AI steering approaches, including our CAA method. For instance, CAA can be used to steer the model towards more harmful, biased, or toxic outputs. We encourage users of this technique to be responsible and avoid increasing harmful behaviors via steering.

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